ZERAVIA | DATA SCIENCE MINI PROJECT

Customer Behaviour Analysis & Purchase Prediction

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1. Business problem

The objective is to explore customer demographics, transaction history and engagement indicators and build a basic model that estimates whether a customer will purchase. The binary target is purchased (1 = yes, 0 = no).

2. Dataset and assumptions

A reproducible synthetic dataset of 1,200 customer records is used for practice. It includes age, income, website visits, email clicks, previous orders, average order value, recency, membership, region and purchase outcome. Missing values and duplicate rows are deliberately introduced to demonstrate cleaning. These are not real client records.

3. Cleaning and transformation

The notebook removes exact duplicate rows, standardises category labels and marks invalid values as missing. It derives age groups, spending bands and an engagement score, then uses Pandas grouping and aggregation. Missing values are imputed within the modelling pipeline.

4. NumPy, EDA and visualisation

NumPy demonstrates mean, percentiles, counts and clipping. Seven charts cover purchase outcome, age distribution, order value by purchase outcome, membership purchase rate, website visits by outcome, numerical correlations and purchase rate by age group. The notebook includes chart titles, axis labels and interpretation guidance.

5. Statistical analysis

Mean, median, standard deviation, variance, minimum and maximum are calculated for key numerical variables. The IQR rule flags potential outliers. Extreme values should be investigated and retained when valid, rather than removed automatically.

6. Machine-learning approach

The primary target is binary, so Logistic Regression is used. An 80/20 stratified train-test split is applied. Numeric features are median-imputed and standardised; categorical features are mode-imputed and one-hot encoded. The preprocessing steps are fitted on training data only through a scikit-learn pipeline. Customer ID is excluded.

7. Evaluation

The notebook reports accuracy, precision, recall, F1-score, ROC-AUC, a classification report, confusion matrix and ROC curve. Linear Regression is not the primary model because purchase outcome is categorical rather than continuous. The actual scores are generated when the notebook runs; they must not be interpreted as real-client performance.

8. Recommendations to test

1) Compare conversion by membership tier. 2) Test follow-up campaigns for recently engaged non-buyers. 3) Compare email engagement and conversion. 4) Evaluate retention offers for customers with prior orders. 5) Investigate extreme order values. 6) Select classification thresholds based on outreach costs. 7) Check performance across segments. 8) Monitor drift and retrain using approved recent data.

9. Limitations and conclusion

Synthetic data is suitable for learning the workflow but cannot support factual claims about a real business. Correlation does not prove causation. For client use, replace the generated data with authorised records, verify that all features are available before the purchase event, assess privacy requirements, compare against a baseline and validate on future-period data.

Deliverables: Jupyter notebook, synthetic CSV dataset, and this concise PDF report.